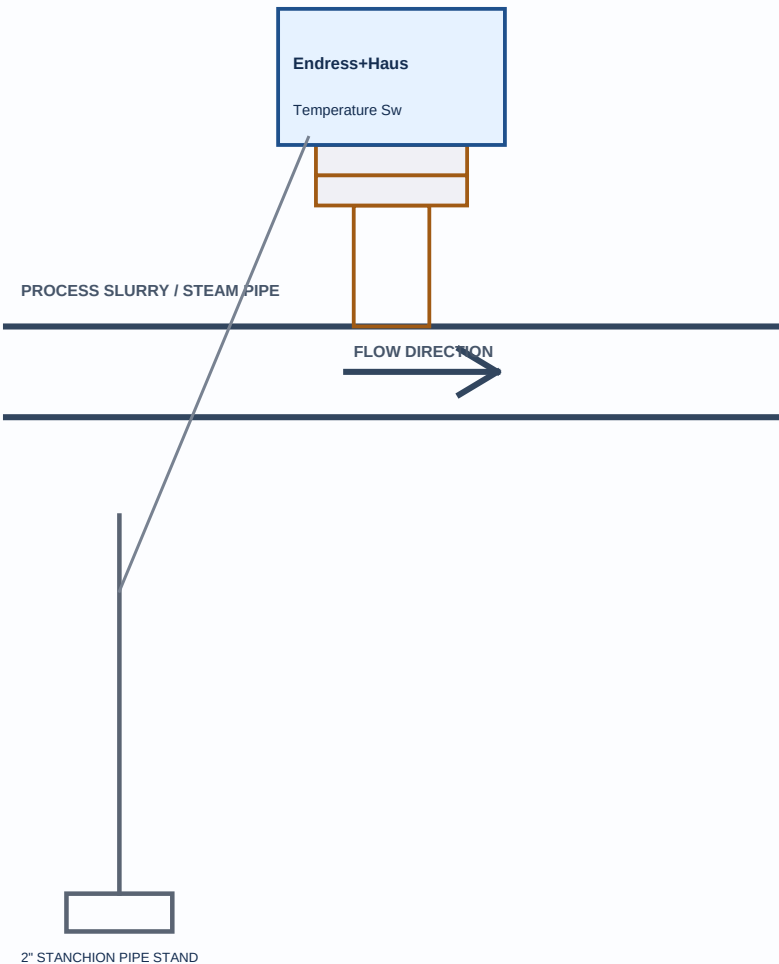


TEMPLATE TYP-22: TEMPERATURE SWITCH

TYPICAL TAGS: TSH-40101 / TSH-60201

SERVICE: High Temperature Safety Trip & Alarm | PLC I/O: 1756-IB32 (24V DC Digital Input)

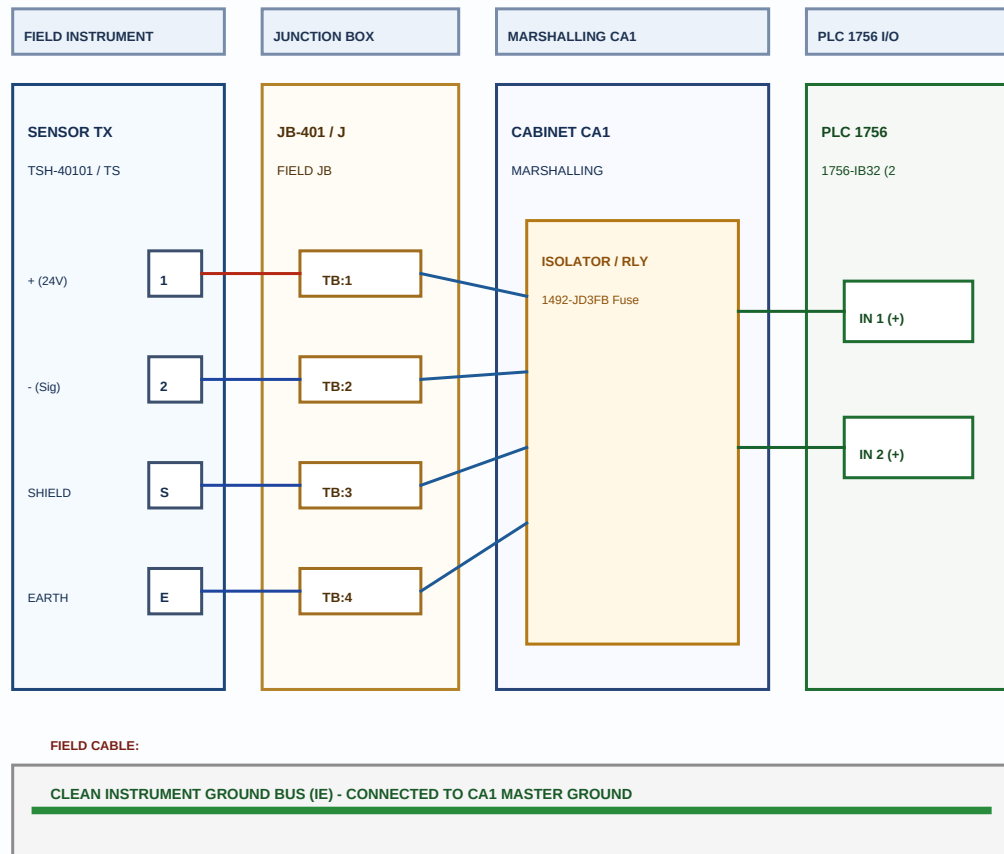
SECTION A: PROCESS PIPING & MECHANICAL HOOK-UP



MECHANICAL HOOK-UP SPECIFICATION

Process Connection:	1/2" NPT Male Threaded / Hyg..
Mounting Type:	Direct Flanged / 2" Pipe Stand
Enclosure Rating:	IP65 / IP67 Stainless Steel ..
Hazardous Class:	General Industrial / Hygieni..
Process Medium:	Starch Slurry / Saturated St..
Ambient Temp Range:	-20°C to +60°C Operating

SECTION B: ELECTRICAL LOOP WIRING SCHEMATIC (FIELD TO PLC)



ELECTRICAL & SIGNAL LOOP SPECIFICATIONS

Signal Protocol:	DI: High Temperature Alarm Trip
Power Supply:	24V DC (10-30V DC, 35mA)
Field Cable Type:	3Cx1.5mm² Shielded Instrument Cable
Destination JB:	JB-401 / JB-602
Marshalling Cabinet:	CA1 / Master Control Panel (Bay M1..M4)
Barrier / Isolation:	1492-JD3FB Fuse Protection Terminal
PLC Module:	1756-IB32 (24V DC Digital Input)

SECTION C: BILL OF MATERIALS (BOM) & NOTES

MANUFACTURER: ENDRESS+HAUSER

MODEL: Thermophant TTR35 (Compact Pt100 Switch)

SERVICE: High Temperature Safety Trip & Alarm

ITEM	MATERIAL / COMPONENT DESCRIPTION	SPEC / MODEL	QTY
1	Thermophant TTR35 Temp Switch E+H Pt100 Class A, Digital Displ		1 EA
2	Barstock Thermowell SS316L (U= 1/2" NPT x 1/2" NPT Process Well		1 EA
3	M12 4-Pin Female Cable Assembl M12 Right Angle Molded Connector		1 EA
4	Thredolet / Boss 1/2" NPT SS3 Pipe Welding Fitting		1 EA
5	Grounding Clamp & Lug Copper Earth Wire		1 SET

ENGINEERING INSTALLATION NOTES

- Configure switch point via optical touch keys on front cover without opening enclosure.
- Normally closed (NC) contact output configuration guarantees fail-safe alarm on wire break.
- Thermowell allows switch replacement without draining the process line.

INGREDION (THAILAND) CO., LTD.
KALASIN STARCH PLANT - JET COOKER PROJECTELECTRONIC TEMPERATURE SWITCH HOOK-UP & LOOP DIAGRAM
INSTRUMENT HOOK-UP & TYPICAL LOOP DIAGRAMDRAWN: AEC-ENG
CHECKED: LEAD-PESTATUS: APPROVED
SCALE: N.T.S.AEC INDUSTRIAL ENGINEERING
SYSTEMS INTEGRATION & AUTOMATIONDWG NO: KAL-JC-TYP-022
REV: 3.6 DATE: 2026-09-08PROJECT NO:
R5THS00138-JCSHEET NO:
24 / 25